Manifold 0.9.1

Version 0.9.1 (build 22) · 2 November 2026 · Beta for testers

Summary

0.9.1 makes SDI output and the window keep better time, and fixes how HDR is shown in the window. SDI and HDMI output no longer depend on your desktop displays: with a 60 Hz main display, 59.94 output no longer stutters, and 1080p119.88 is frame-accurate. Each window now follows the display it is on. HLG and PQ look right in both display modes: As macOS Shows It matches QuickTime, and HLG in Reference is no longer far too bright. iPhone 120 fps clips play, the control bar fits narrow and vertical windows, and the Sync and Color Guide is now in the Help menu.

Highlights

- **SDI no longer stutters when your main display runs at 60 Hz.** 59.94 output used to repeat and skip a frame every few seconds with a 60 Hz main display. Output is now timed by its own frame rate, not by a desktop display.
- **HDR in As macOS Shows It matches QuickTime.** HLG and PQ are handed to macOS's own HDR handling, so bright highlights roll off instead of clipping to white.
- **HLG in Reference is fixed.** It was shown far too bright in 0.9.0. It now follows the broadcast standard, with HLG's reference white at 203 nits.
- **Saved streams don't need recalibrating.** Your saved A/V offsets carry over and still hold.

New

COLOR

- **Dolby Vision in the Inspector.** A file that carries Dolby Vision now says so, with its profile, its level and what Manifold shows. For example: "Profile 8.4 (level 7) · HLG base layer shown; DV metadata not applied". Manifold shows the standard picture inside the file and does not apply Dolby Vision itself.
- **A warning for Dolby Vision files Manifold can't show.** A profile 5 file (common for streaming masters), or any Dolby Vision file with no standard picture inside, now puts a warning across the top of the window: Manifold can't display it correctly. The Inspector says the same. Close the warning and it stays closed until you open another file. Profile 8 and 7 files, whose standard picture Manifold shows correctly, get no warning.

INTERFACE AND SHORTCUTS

- **Help ▸ Sync and Color Guide.** The guide to live stream sync and color management is now built into Manifold, so it opens offline and always matches the version you have. It follows light and dark mode, and you can search it with ⌘F.

Improved

COLOR

- **HLG in As macOS Shows It.** Manifold now hands an HLG picture to macOS's HLG adjustment, as QuickTime does. On a display in SDR mode, an iPhone HLG clip no longer clips faces, white shirts and sky to white. With HDR on, the picture matches QuickTime by eye.
- **PQ in As macOS Shows It.** Manifold now hands a PQ picture to the macOS HDR10 tone map, as QuickTime does. It uses the file's mastering brightness when the file declares it, and macOS's defaults when it doesn't. On a display in SDR mode, PQ was about twice as bright as QuickTime and clipped everything from 100 nits up. It now matches QuickTime on every grey step. With HDR on, it matches QuickTime by eye.
- **This follows the picture, not just the source.** A LUT whose output is set to PQ or HLG is treated the same way. A PQ or HLG source under a LUT that outputs Rec. 709 is not.
- **The scopes still show the file.** In As macOS Shows It, macOS now adapts HLG and PQ to your display, so above the display's limit the window and the scopes no longer agree. The picture also changes with the display's HDR headroom and, for HLG, its brightness. The transform chain readout says so, for example "macOS HLG adjustment · follows display brightness · scopes show the file".
- **Reference is absolute for HDR.** In Reference, PQ and HLG are shown as absolute light: 1.0 on the display is 100 nits, with no tone mapping. Above what the display can show, the picture clips. On a display in SDR mode, the readout now says "display in SDR mode · above 100 nits clips", for PQ and HLG alike. PQ in Reference looks the same as in 0.9.0. SDR in Reference is unchanged (BT.1886).
- **Hold to compare means more for HDR.** As macOS Shows It and Reference now differ for HLG and PQ, so holding C shows what macOS's tone mapping does to the picture.
- **The HDR changes are in the window only.** SDI and HDMI color, the HDR signal sent with them, the scopes and exported frames are as in 0.9.0 in every display mode.

SDI/HDMI OUTPUT

- **SDI and HDMI output no longer depend on your desktop displays.** While output runs, frames are sent on a timing set by the output's frame rate, not by the display the window is on. A 59.94 output played every frame, whichever display the window was on and when it was dragged between them. A ten-minute run played 36,000 frames with none late, dropped or repeated.
- **SDI output no longer stutters when your main display runs at 60 Hz.** With a 60 Hz main display, 59.94 output used to repeat and skip a frame every few seconds. That is gone.
- **1080p119.88 output is now frame-accurate.** It used to repeat and skip about three frames a second.
- **Live streams to SDI are steadier.** A live stream sent to SDI no longer repeats or skips frames at the card (it used to, several a second).

Fixed

COLOR

- **HLG in Reference was far too bright** (a bug in 0.9.0 and earlier). HLG at 50 % was shown at SDR white, and everything above it clipped. Reference now converts HLG to PQ as the broadcast standard (BT.2408) says, at 1000 nits. HLG's reference white (75 %) now lands at 203 nits, the same as PQ's. On a display in SDR mode, clipping now starts at about 63 %, as a PQ picture clips at 100 nits.
- **Reference's display calibration settings are greyed for HDR pictures.** In Reference, the This display section (the checkbox, Display white and Display black) applies only to SDR. For a PQ or HLG picture, including one set by a tag override or a LUT, it was editable but did nothing. It's now greyed, and says so. What you set comes back into use for the next SDR source. The labels also now say what they ask for: "This display is calibrated to BT.1886 (Rec. 709)", "Display white (cd/m²)" and "Display black (cd/m²)".
- **Switching display modes changes the picture in one step.** The new mode's color settings and the repainted picture now reach the screen together. Before, they were sent a few milliseconds apart, so for a moment the old picture could be shown with the new mode's settings.
- **A false warning on every display-mode change is gone** (it only showed in diagnostics). Diagnostics reported that a frame had been shown with the wrong color each time the display mode or a color override changed. None was: the warning was counting the wrong thing. It still reports a real late color change when a file opens.

LIVE STREAMS

- **Install NDI Runtime… could fail to download.** NDI's short link stopped working. The button and the menu item now download the installer directly, and a link to NDI's own downloads page sits beside them in case the file ever moves.

FILES AND INTERLACED

- **iPhone 120 fps clips and other high-frame-rate files now play** (a bug in 0.9.0 and earlier). At about 96 fps and above, the picture froze on the first frame, or stuttered, while the audio played on. Files at 96, 100, 119.88 and 120 fps, including iPhone slow-motion clips, now play at their own frame rate, or at your display's refresh rate if the file is faster.
- **Confirmed on a MacBook Air:** the iPhone rotation fix in 0.9.0 build 21 works. Clips open the right way up, and a vertical window can be made smaller.

INTERFACE AND SHORTCUTS

- **The control bar fits narrow and vertical windows** (a limitation in 0.9.0 and earlier). In a window too narrow for the full bar, controls at both ends, Play among them, were cut off. The bar now changes shape to fit, in the floating bar and the docked bar alike:
- **Compact:** the less-used tools (Edit in Flip, Refresh Metadata, subtitles, Export Frame, framing guides and Loop) move into a **More (…)** menu, and the color button shows a short label such as "709" or "2020 PQ".
- **Two rows** in the narrowest windows, such as a vertical clip at its smallest: playback on the first row, output and color on the second.
- **More (…) lights up** when anything inside it is on, such as Loop, subtitles or a framing guide.

- **The window-size readout is placed for narrow windows.** It now sits at the top center, over the picture, and below the file name when that is shown. If it doesn't fit on one line it wraps instead of being cut off. The readout and the Inspector no longer sit partly outside a narrow window.
- **Window positions no longer pile up in Manifold's settings.** Every update saved the main window's position under a new name and never read the old ones. It now saves it under one fixed name that stays the same from update to update.

LICENSING AND PRIVACY

- **About Manifold ▸ Licences now gives NDI's licensing page.** The links inside NDI's own license notice use NDI's short links, which have been down since October 29. The notice still ships exactly as NDI wrote it; About now says so and gives NDI's licensing page instead. The NDI SDK License Agreement itself is included in the NDI SDK download.

PERFORMANCE

- **The picture keeps time with the display the window is on.** With more than one display, every window used to update at the main display's refresh rate wherever it was. A window on a 60 Hz display was redrawn 120 times a second when the main display ran at 120 Hz, and a window on a faster display was held to the main display's rate. Each window now follows its own display, including when you drag it from one display to another while it plays.
- **A live stream at 59.94 now plays smoothly on a 60 Hz display.** It could show about 50 new pictures a second instead of 60, in bursts. The window now chooses each picture for the moment it reaches the screen, which also brings the window's picture slightly closer to the sound.
- **High-frame-rate files keep their full rate in the window during SDI or HDMI output.** On a 120 Hz display, a 100 or 119.88 fps file used to update 60 to 100 times a second while output ran. It now shows every frame. The output itself was, and is, frame-accurate.

Known limitations

COLOR

- **Very saturated PQ colors in As macOS Shows It differ slightly from QuickTime.** Colors far outside your display's range, such as a bright Rec. 2020 green or cyan, can look a little brighter in Manifold than in QuickTime. Colors your display can show match.
- PQ or HLG on primaries other than Rec. 2020 or P3-D65 can't be shown as stated, and those overrides are greyed out (as in 0.9.0).

SDI/HDMI OUTPUT

- **During long playback from a file, SDI output can repeat or skip a single frame every few minutes,** as the output card's clock and your Mac's drift apart (as in 0.9.0).
- **2160p100 and 2160p119.88 output arrive late at the card on every frame** (as in 0.9.0). Use a 1080p mode for these rates.
- A file's picture and sound reach SDI and HDMI about 11 ms later than in 0.8.4 (as in 0.9.0). They stay together on the card; if you line SDI up against another source, check the offset again.
- Over HDMI, an HLG picture goes out without its HLG signal, so the display isn't told to switch to HLG. SDI carries HLG.

- SDI audio goes silent for up to about a tenth of a second at each seek, scrub release and loop point during file playback.

LIVE STREAMS

- Interlaced streams (SRT, WHEP, HLS, NDI) still go out over SDI as progressive, with both fields woven into each frame. An interlaced NDI source arrives already deinterlaced by NDI.
- After a network stall of a second or more, live sound can stay 10–20 ms late for up to a minute.
- OBS stamps 23.976 fps WHIP video very slightly fast. Through MediaMTX on its default settings, sound and picture drift apart by about a quarter of a second an hour over WHEP. Set `useAbsoluteTimestamp: true` on the MediaMTX path to fix it. Cloudflare Stream corrects it on its own.

FILES AND INTERLACED

- Interlaced files that aren't 1080 lines, and 1080 interlaced at 23.98 or 24, have no interlaced output mode. They go out progressive at their own size, and the DeckLink menu says so.
- On 8-bit H.264 files macOS's decoder still blends some color between neighbouring lines before Manifold sees the picture, and on interlaced MPEG-2 it blends the fields completely.

Upgrading from 0.9.0

Nothing moves when you upgrade. Your license, saved streams and settings carry over unchanged.

- **Saved streams don't need recalibrating.** A saved A/V offset doesn't depend on how the window reaches your display, which is what changed, so a sync check on a saved stream reads the same as before. In a window, the picture now arrives a little earlier against its sound than in 0.9.0, by about 26 ms on a 60 Hz display. SDI and HDMI output timing is unchanged.
- **Your main window's saved position is reset once** when you first open 0.9.1, and kept from then on.

For testers: what to look at

- **SDI with a 60 Hz main display.** Play a long 59.94 file out to SDI with your main display at 60 Hz, and with the window on another display. It should play without the repeats and skips 0.9.0 had.
- **Windows on more than one display.** Drag a playing window between displays with different refresh rates. Playback should stay smooth on each.
- **HDR in As macOS Shows It against QuickTime,** with HLG (including iPhone clips) and PQ, on a display in SDR mode and with HDR on. Does Manifold look like QuickTime?
- **HLG in Reference.** Highlights should no longer be blown out the way they were in 0.9.0. On an HDR display, tell us where the brightest parts start to clip.
- **Hold C on HDR files.** Switching between As macOS Shows It and Reference should be instant, with no flash of the wrong picture.
- **High-frame-rate files,** such as iPhone 120 fps clips: the picture should move with the sound from the first frame, with no freeze or stutter.
- **Narrow and vertical windows:** make a window as small as it goes, floating and docked. Every control should stay on screen and reachable, either on the bar or in More (…).

- **Dolby Vision files:** does the Inspector name the profile, and does a profile 5 file show the warning?

Reporting a problem: Help ▸ Export Diagnostics… saves a report we can read (no license details, and no file names unless you tick the box). For a picture problem, Help ▸ Export Frame for Support… saves the frame itself.

About this tester build: it carries extra diagnostics. There is a Debug menu, but it stays hidden unless we ask you to turn it on: one of its items replaces the program audio with a test tone. A few Control-Option key combinations are test shortcuts and connect on their own: ⌃⌥D and ⌃⌥H connect to your first saved SRT or WHEP stream, and ⌃⌥N to the first NDI source on your network. Please avoid them.